

Arrhythmias — Sinus Tachycardia



1. Clockmaker shop

WHAT YOU SEE

An old clockmaker's shop, walls covered in ticking clocks, sign 'CLOCKMAKER' — the master clock that sets the beat is the SA node, the heart's natural pacemaker keeping perfect time.

WHAT IT ENCODES

The sinus (SA) node is the heart's dominant pacemaker — the 'clockmaker.' Sinus tachycardia is a fast (>100 bpm) but otherwise completely normal rhythm: every QRS is preceded by a normal sinus P wave with a constant PR interval. It is a regular, narrow-complex tachycardia that almost always represents a physiologic RESPONSE to a stimulus rather than a primary electrical disease.

BOARD TRAP

Sinus tachycardia is rarely a primary diagnosis — calling fast sinus rhythm a 'disease to be rate-controlled' misses that it is usually a clue to an underlying problem (fever, hypovolemia, pain, sepsis, PE, hyperthyroidism).

2. P axis normal

WHAT YOU SEE

ECG note: up in I, II, III, aVF; down in aVR

WHAT IT ENCODES

True sinus P waves are upright (positive) in leads I, II, and the inferior leads (III, aVF) and inverted in aVR, reflecting the normal high-to-low, right-to-left atrial activation vector from the SA node. Confirming this P-wave axis is how you prove the rhythm is sinus and not an ectopic atrial or junctional tachycardia.

BOARD TRAP

An inverted P wave in lead II (or a P axis that is abnormal/retrograde) means the rhythm is NOT sinus — think low atrial focus or junctional/AVNRT rhythm. Assuming any fast narrow-complex rhythm is 'sinus tach' without checking P-wave axis is a classic error.

3. Metronome SVT

WHAT YOU SEE

A metronome ticking steadily, labelled 'SVT' — the rock-steady, even tick = the regular, narrow-complex rhythm of sinus tach, one P wave before every beat.

WHAT IT ENCODES

Sinus tachycardia is a regular, narrow-complex (QRS <120 ms) rhythm with a clearly visible P wave before every QRS — a steady metronome. Rate in adults is typically 100-150 bpm, rarely exceeding ~150-160 even with maximal stress (max sinus rate ≈ 220 minus age).

BOARD TRAP

If the regular narrow-complex rate is ~150 bpm, suspect ATRIAL FLUTTER with 2:1 block (atrial rate ~300/2) masquerading as sinus tach — look for sawtooth flutter waves. A 'sinus tach' that won't go below 150 is a red flag for flutter.

4. Gradual onset

WHAT YOU SEE

A blood-bag IV drip with a 'GRADUAL' label, fluid building up slowly — sinus tach ramps up and winds down gradually as autonomic tone shifts, never switching on like a light.

WHAT IT ENCODES

Sinus tachycardia 'warms up' and 'cools down' GRADUALLY as autonomic tone changes — the rate ramps over seconds to minutes. This gradual onset/offset is the key bedside discriminator from re-entrant PSVT.

BOARD TRAP

Abrupt, instantaneous start/stop is the wrong descriptor for sinus tachycardia — sudden on/off points to a re-entrant PSVT (AVNRT/AVRT). Describing sinus tach as 'paroxysmal' is incorrect.

5. Abrupt = PSVT

WHAT YOU SEE

A lightning bolt striking with an 'ABRUPT' label — the sudden flash = the instant on/off of re-entrant PSVT (AVNRT/AVRT), the opposite of sinus tach's slow ramp.

WHAT IT ENCODES

Abrupt (sudden) onset and abrupt termination characterize PAROXYSMAL re-entrant SVT (AVNRT, AVRT) — the patient often says it started and stopped 'like a switch,' frequently with palpitations and a rate of 150-250 bpm. This contrasts directly with sinus tachycardia's gradual ramp.

BOARD TRAP

Vagal maneuvers/adenosine TERMINATE re-entrant PSVT but only transiently SLOW sinus tachycardia (which resumes) — so a tachycardia that abruptly converts to sinus with adenosine was PSVT, not sinus tach.

6. Treat the cause

WHAT YOU SEE

A pennant reading 'TREAT THE CAUSE' — sinus tach is a response, so the banner tells you to chase the underlying driver, not just slow the rate.

WHAT IT ENCODES

Because sinus tachycardia is a compensatory response, management is to FIND and TREAT the underlying driver — give fluids for hypovolemia, antipyretics for fever, oxygen/anticoagulation for PE, antithyroid therapy for thyrotoxicosis, transfuse/treat anemia, control pain and anxiety.

BOARD TRAP

Giving a beta-blocker or other rate-control drug WITHOUT treating the cause is the wrong first step and can be dangerous — blunting compensatory tachycardia in a hypovolemic or septic patient removes a vital compensatory mechanism and can precipitate decompensation/shock.

7. Thyroid butterfly

WHAT YOU SEE

A butterfly (thyroid gland) spread over a face — the butterfly-shaped thyroid in overdrive (hyperthyroidism) revs the heart, a reversible cause of sinus tach.

WHAT IT ENCODES

Hyperthyroidism (thyrotoxicosis) is a key reversible endocrine cause of sinus tachycardia — excess thyroid hormone upregulates beta-adrenergic responsiveness. Check TSH/free T4 in unexplained persistent sinus tach; thyrotoxicosis is also a major precipitant of atrial fibrillation.

BOARD TRAP

In a thyrotoxic patient, beta-blockers (propranolol) help symptomatically but you must still treat the thyroid itself — and avoid amiodarone, which is iodine-rich and can worsen thyroid dysfunction.

8. Anemia

WHAT YOU SEE

A pale, washed-out face labelled 'ANEMIA' — the pallor = low oxygen-carrying blood, forcing the heart to beat faster to compensate.

WHAT IT ENCODES

Anemia lowers oxygen-carrying capacity, so the heart compensates by raising cardiac output via a higher heart rate — high-output compensatory sinus tachycardia. It is a common, easily checked cause (CBC) of unexplained tachycardia.

BOARD TRAP

Rate-controlling an anemic patient's compensatory sinus tachycardia instead of correcting the anemia (transfusion/iron) treats the number, not the patient, and worsens tissue oxygen delivery.

9. Pulmonary embolism

WHAT YOU SEE

Lungs labelled 'PULMONARY EMBOLISM' with an 'EXCLUDE FIRST' tag — the can't-miss lung clot you must rule out first in unexplained sinus tach.

WHAT IT ENCODES

PE is the can't-miss cause of unexplained sinus tachycardia — it is the most common ECG finding in PE. Look for supporting clues: hypoxia, pleuritic chest pain, the S1Q3T3 pattern, or new right-heart strain (RBBB, T-wave inversions V1-V4).

BOARD TRAP

Attributing unexplained sinus tachycardia to 'anxiety' without excluding PE (and other dangerous causes) is a dangerous anchoring error — always exclude PE in the acute, unexplained setting.

10. Resting tachy

WHAT YOU SEE

A speedometer-style 'REST' gauge with the needle pinned high — the heart races even at rest, the picture of inappropriate sinus tachycardia (IST).

WHAT IT ENCODES

Inappropriate sinus tachycardia (IST) is a persistently elevated resting sinus rate (>100 bpm at rest, mean 24-h HR >90) that is disproportionate to physiologic demand, with a normal P-wave axis. It is a diagnosis of EXCLUSION after ruling out secondary causes (PE, anemia, hyperthyroid, fever, drugs, deconditioning).

BOARD TRAP

IST is diagnosed only AFTER excluding all secondary causes — labeling a patient with IST while a treatable driver (e.g., hyperthyroidism, anemia) is present is wrong.

11. ≥30 bpm on standing

WHAT YOU SEE

A figure standing up beside '≥30 bpm' and a 'NO BP DROP' note — standing jumps the heart rate ≥30 with no fall in blood pressure, the defining combo of POTS.

WHAT IT ENCODES

POTS (postural orthostatic tachycardia syndrome) = a sustained HR rise of ≥30 bpm (≥40 bpm in adolescents) within 10 minutes of standing, WITHOUT a significant orthostatic blood-pressure drop, plus orthostatic symptoms. First-line management is non-pharmacologic: increased salt and fluid intake, compression stockings, and exercise reconditioning; ivabradine or low-dose beta-blockers are added if needed.

BOARD TRAP

The discriminator is the BP: a fall in BP on standing (≥20/10 mmHg) = orthostatic HYPOTENSION, not POTS. POTS specifically requires the tachycardia WITHOUT the BP drop.

12. SN ablation last

WHAT YOU SEE

A scale balancing 'SN ablation' against BP meds — ablation sits at the bottom of the ladder, weighed against (and after) drugs because it risks needing a pacemaker.

WHAT IT ENCODES

For inappropriate sinus tachycardia, the therapeutic ladder is: lifestyle/hydration first, then ivabradine (a selective I-f 'funny current' SA-node blocker that slows rate without lowering BP) or beta-blockers; sinus node modification/ablation is a last resort given its risk of needing a permanent pacemaker.

BOARD TRAP

Sinus node ablation is NOT first-line for sinus tachycardia (appropriate or inappropriate) — jumping to ablation before drugs/lifestyle, or ablating a SECONDARY sinus tach, is wrong and risks creating pacemaker dependence.

13. Benign post-viral

WHAT YOU SEE

A green 'BENIGN' check beside '3-12 MONTHS POST-VIRAL' — the green tick marks post-viral sinus tach as usually harmless, self-resolving over 3-12 months.

WHAT IT ENCODES

Post-viral inappropriate sinus tachycardia (including post-COVID dysautonomia) is frequently benign and self-limited, resolving over roughly 3-12 months. Reassurance plus symptomatic measures is usually sufficient.

BOARD TRAP

Even when post-viral sinus tach is presumed benign, you must still exclude structural disease and other secondary causes before reassurance — and avoid aggressive ablation for a condition that typically self-resolves.